

K1401 — corrections round + the Cal/Lyra reconciliation. ★ Grace + Cal caught FOUR things, TWO of which correct my own K1399/K1400 leads: (1) the Verlinde-137 ($\dim V_7 = 964M$) is an UNVERIFIED corpus claim (BST_WindingToZeta), NOT banked — the banked, verified Verlinde prime is 1747 (genus-3), a DIFFERENT number. RETRACT my Verlinde-137 corroboration until the genus-7 Verlinde formula is actually computed. (2) 137 is a RATIONAL prime $\equiv 1 \pmod{4}$ that SPLITS in $\mathbb{Z}[i]$ as $(11 \pm 4i)$ — it is NOT itself a Gaussian prime (the factors $11 \pm 4i$ are); the irreducibility that kills decompositions is *rational* primeness. Correct my “Gaussian prime” wording. (3) the SPECTRAL CAP is IMPOSED, not derived — RealityBudget admits “computing the formal degrees explicitly has not been carried out”; 137 is *inserted* as the cap. (4) T1286 “unique prime” overreaches (149 also closes the loop). ★ ADJUDICATION of the Cal/Lyra tension: the cap-forcing can proceed as mode-count = bulk ($N_c^3 \cdot n_C$) + boundary (rank) IFF each piece is target-innocent — reconciles “no assembling a prime” (Cal) with “two forced pieces” (Lyra). Cal cold-reads.

Keeper (2026-08-12, ~10:45. The team stress-tested the geometry — including my leads — and it’s the honest kind of round: three corrections that tighten the claim, one that corrects me, and a real Cal/Lyra tension to settle. Cal cold-reads. Nothing pushed.)

★ Corrections to MY OWN leads (K1399/K1400) — ratified, retracted

- Verlinde-137 → RETRACTED as banked corroboration. I cited “137 | $\dim V_7 = 964,141,747$ ” from BST_WindingToZeta. Grace: that number is not verified as an actual Verlinde dimension — it appears in one note but was never confirmed by computing the Verlinde formula at genus 7. The banked, verified Verlinde prime is 1747 (genus-3: $\dim V_3 = n_C \cdot g^3 + 2^{n_C}$), a *different* number. So: the Verlinde *route* is real evidence this geometry produces indivisible conformal blocks — but the specific “137 divides a Verlinde dimension” claim is UNVERIFIED and must not be cited until the genus-7 Verlinde formula is computed. My K1400 caveated the *specialness*; Grace found the *number itself* is unconfirmed. Retracted to “unverified corpus claim; the verified Verlinde prime is 1747, not 137.” (This is the walk-back rate staying healthy through the auditor’s chair — twice on the same lead.)
- “Gaussian prime” → corrected to “rational prime $\equiv 1 \pmod{4}$, splits in $\mathbb{Z}[i]$ ” (Cal). $137 = (11+4i)(11-4i)$ *splits* in the Gaussian integers — so 137 is NOT a Gaussian prime; the factors $11 \pm 4i$ are. The irreducibility that kills the decomposition attempts is that 137 is a RATIONAL prime (irreducible in $\mathbb{Z}$) — that’s the load-bearing point. Say: “ $137 = 11^2 + \text{rank}^4$ is the unique two-square form of a rational prime $\equiv 1 \pmod{4}$, which splits in $\mathbb{Z}[i]$; rank^4 sits inside it.” Correct my K1399/K1400 wording.

★ Grace’s two other corrections (ratified)

- The SPECTRAL CAP is IMPOSED, not derived — and the corpus admits it in its own words: RealityBudget’s proof note says the Haldane truncation at $N_{\max}=137$ “adds a non-standard element” (Plancherel normally sums over all modes) and “*computing the formal degrees explicitly has not been carried out.*” So 137 is INSERTED as the cap, not computed to be the cap. It’s NOT circular in the $[1/\alpha]$ sense (never touches α) — but it is NOT forced from the geometry either. This is the honest state and the exact open computation: derive the cap from the domain, α -free. (Confirms Grace’s earlier flag; it’s the one computation the corpus wrote down and never carried out.)

- T1286 “unique prime” overreaches (flagged to me, ratified): 137 closes the self-reference smooth-counting loop (through $54 = \text{rank} \cdot N_c^3$ and $135 = N_c^3 \cdot n_C$), but 149 closes the same loop too — 137 is only singled out by the *softer* “intermediates are clean BST integers” criterion. So it corroborates but does not uniquely force; the “unique prime” label is retired to “closes the loop with clean BST intermediates (non-unique).”

★ The Cal/Lyra tension — reconciled (the load-bearing adjudication)

- Cal: “ $137 = 135 + 2 \rightarrow \text{REJECT}$; a rational prime has no forced decomposition; K38’s +rank is corroboration, not the α -forcing; don’t present the decomposition as the derivation.”
- Lyra: “two-thirds done — the +rank is forced three ways (K38); what’s left is showing $135 = N_c^3 \cdot n_C$ is the size of the space of bound modes (a counting job).”
- ★ RECONCILIATION: both are right, and the discipline that separates them is target-innocence of each piece. Cal’s rejection is correct for *assembling* — picking “ $135 + 2$ ” *because* it sums to 137. It does NOT forbid *two independently-forced geometric mode-counts that happen to total a prime*. The primality of the sum is then a *consequence* (and corroboration — it’s irreducible, fitting a fundamental cutoff), not the mechanism. So the cap-forcing PASSES iff:
 - (a) the “135” is DERIVED as the count of bulk bound modes from the domain structure, target-innocent (NOT “ $137 - 2$ ”); AND
 - (b) the “+rank” is the K38-forced boundary/Hodge pair (already target-innocent, 3 routes, 93%).
 - Then the forced sum $N_c^3 \cdot n_C + \text{rank} = 137$ is a derivation, and it’s NOT “decomposing a prime” — it’s counting two forced mode-types that total a rational prime. The guard (Cal’s): each piece target-innocent; reject any “135” that is secretly “ $137 - 2$ ”.
- ☐ Lyra’s “two-thirds done” is DEFENSIBLE *if* the 135 is derived target-innocent; Cal’s caution is the guard on it. The remaining forcing = derive $135 = N_c^3 \cdot n_C$ as the bulk bound-mode count, α -free and target-innocent.

Bank Elie’s structural finding (corroboration, not forcing)

$137 = 11^2 + \text{rank}^4 = 121 + (\text{store-16})$. The 16 the boundary stores (the operator-space = rank^4) is *literally* the rank^4 term in 137’s unique two-square form. So the day’s two forcings — store-16 and force-137 — share the number 16, α -free and target-innocent. Doesn’t force anything, but it’s real structure corroborating the “one knot” from the number-theory side (the same 16 in the boundary’s storage and in the prime’s irreducible form). Note it as corroboration.

The honest net (calibrate both directions)

- 137 is real geometry, not a fit — several α -free FINGERPRINTS: the Fermat form with rank^4 (verified), the Verlinde-1747 primality (banked; the 137-Verlinde is unverified), the near-fixed-point (non-unique), Elie’s $137 = 11^2 + \text{store-16}$. Together they say plainly it’s an invariant, not a coincidence. That’s worth a lot — the difference between a coincidence and an invariant.
- BUT there is still NO α -free derivation that the spectral truncation MUST be 137. Corroboration $\neq$ forcing. The spectral cap is imposed (corpus admits it). The last stone = the one computation the corpus wrote down and never did: derive the cap α -free, via the target-innocent bulk-mode count (135) + the K38 boundary pair (+2).

Route

1. ★ Grace + Lyra — the cap-forcing, α -free + target-innocent: derive $135 = N_c^3 \cdot n_C$ as the count of bulk bound modes from the domain structure (the formal degrees the corpus never computed — RealityBudget) — NOT as “ $137 - 2$ ”; combine with the K38-forced +rank. If both target-innocent, the forced sum $137 =$ the cap, α -free $\rightarrow \alpha = 1/137$. Reject $[1/\alpha]$; reject any “135” that’s secretly “ $137 - 2$ ”.
2. Grace — the Verlinde homework (corroboration): compute the actual genus-7 Verlinde dimension (is 964M real? does 137 divide it? is it special?) — OR rest on the banked genus-3 1747. Do NOT cite the 137-Verlinde until computed.

3. Data-layer corrections (Grace lane, Keeper-ruled): T1286 “unique prime” → “closes the loop, non-unique”; “Gaussian prime” → “rational prime $\equiv 1 \pmod{4}$, splits in $\mathbb{Z}[i]$ ” wherever it appears.
4. Cal — hold the target-innocence guard on each mode-count piece (135 not “137–2”); the α -free bar; store-16 stamped.
5. Keeper — ratified the corrections (incl. retracting my own Verlinde-137 + Gaussian-prime); adjudicated Cal/Lyra (target-innocent pieces); hold the tier (fingerprints $\neq$ forcing; cap-forcing is the open computation); recused.

— Keeper, K1401, 2026-08-12. Corrections round. RETRACT my Verlinde-137 (dim $V_7=964M$ is an UNVERIFIED corpus claim, BST_WindingToZeta; banked Verlinde prime = 1747 at $g=3$, different number — don’t cite 137-Verlinde until genus-7 formula computed). CORRECT “Gaussian prime” → “rational prime $\equiv 1 \pmod{4}$, splits in $\mathbb{Z}[i]$ as $11 \pm 4i$ ” (Cal; the irreducibility is RATIONAL primeness). Grace: SPECTRAL CAP is IMPOSED not derived (RealityBudget admits “formal degrees not carried out”; 137 inserted, not computed; not circular but not forced) — THE open computation. T1286 “unique prime” overreaches (149 also closes the loop; non-unique, soft BST-integer criterion). CAL/LYRA RECONCILIATION: cap-forcing = mode-count bulk($N_c^3 \cdot n_C=135$) + boundary(rank) PASSES iff EACH piece target-innocent (135 derived as bulk bound-mode count, NOT “137–2”; +rank = K38 3-route Hodge pair). Reconciles “no assembling a prime” (Cal) with “two forced pieces” (Lyra); primality = consequence + corroboration, NOT mechanism; guard = each piece target-innocent. Elie’s finding (corroboration): $137 = 11^2 + \text{rank}^4 = 121 + \text{store-16}$ — the two forcings share the 16, α -free. NET: 137 real geometry (α -free fingerprints: Fermat/rank⁴, Verlinde-1747, near-fixed-point, $137=11^2+\text{store16}$) but NO α -free derivation the cap MUST be 137 — corroboration $\neq$ forcing; the open stone = derive the cap α -free via target-innocent 135 + K38 +2. Route: Grace/Lyra force the cap (135 as bulk-mode-count target-innocent, α -free); Grace verify genus-7 Verlinde or rest on 1747; data-layer corrections (T1286, Gaussian-prime); Cal target-innocence guard; Keeper ratified+adjudicated+retracted-own-leads, recused. Cal cold-reads. Nothing pushed.